

# PROJECT CHARTER

## Supply Chain Risk & Performance Intelligence Platform

|               |                                            |                  |                                    |
|---------------|--------------------------------------------|------------------|------------------------------------|
| Project Team: | Data & AI Engineering Group (2-Person MVP) | Date:            | July 1, 2026                       |
| Focus Area:   | Data Engineering, ML, & Analytics          | Target Industry: | Fashion & Beauty Retail Operations |

### 1. PROJECT OVERVIEW & BUSINESS PROBLEM

Fashion and beauty organizations navigate highly complex, volatile global supply chains. Operational inefficiencies heavily degrade margins, customer satisfaction, and agility. Traditional e-commerce analytic workflows lack the operational context required to diagnose structural pipeline flaws.

This project establishes a comprehensive, end-to-end **Supply Chain Risk & Performance Intelligence Platform**. The solution is specifically engineered to surface real-time granular variables—such as varying lead times, supplier reliability thresholds, transportation modes, and defect rates—to eliminate five critical business pain points:

- **Stockouts:** Revenue leakage resulting from product unavailability during peak demand cycles.
- **High Shipping Costs:** Inflated logistics expenses and inefficient carrier selection.
- **Supplier Delays:** Lack of predictability regarding component and finished-goods arrivals.
- **Manufacturing Defects:** Margin degradation due to low quality-assurance pass rates at production plants.
- **Poor Inventory Planning:** Capital tied up in slow-moving stock or lost to unoptimized allocation.

**Strategic Objective:** Transform raw operational supply chain metrics into localized data models and predictive infrastructure. This enables executive and logistics teams to identify operational failure modes proactively before they disrupt production or fulfillment channels.

### 2. PROJECT SCOPE & CROSS-DISCIPLINARY ARCHITECTURE

The platform unifies three core modern data competencies—Data Engineering, Machine Learning, and Business Intelligence—into a singular infrastructure stack designed around a 4-week Minimum Viable Product (MVP) delivery schedule.



| DATA ENGINEERING                                                                                                                                                                                                                                                       | MACHINE LEARNING                                                                                                                                                                                                                 | ANALYTICS & BI                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Profiling raw transactional data.</li><li>• Constructing scalable ETL pipelines.</li><li>• Maintaining data quality constraints.</li><li>• Establishing a medallion architecture warehouse (Bronze → Silver → Gold).</li></ul> | <ul style="list-style-type: none"><li>• Feature engineering from analytical layers.</li><li>• Constructing predictive models for risk scoring.</li><li>• Single-model prioritization targeting stockout probabilities.</li></ul> | <ul style="list-style-type: none"><li>• Translating dimensional star schemas into dynamic operational semantic models.</li><li>• Publishing specialized dashboard structures for corporate stakeholders.</li></ul> |

### 3. TECHNICAL DELIVERABLES & SPECIFICATIONS

#### 3.1 Data Engineering: Medallion Architecture & Star Schema Design

Data pipelines ingest diverse multi-source metrics (manufacturing volumes, shipping legs, lead periods) into a formalized Medallion data structure:

- **Bronze Layer (Raw):** Append-only preservation of source datasets exactly as received to guarantee traceability and auditability.
- **Silver Layer (Clean):** Data cleansing scripts dedicated to removing structural duplicates, resolving missing elements via contextual imputation, standardizing geographic naming conventions, validating economic limits, and converting operational periods to precise numerical vectors.
- **Gold Layer (Analytics):** A highly structured star schema optimized for fast analytical query patterns.

| FACT TABLE: FACTSUPPLYCHAIN                                                                                                                                                                                                                                                                                                                                                                              | DIMENSION TABLES                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Core Granularity:</b> Individual SKU per Transactional Event</p> <ul style="list-style-type: none"><li>• SKU (FK)</li><li>• Revenue Generated</li><li>• Total Operational Costs (Manufacturing + Logistics)</li><li>• Lead Times (Actual vs. Expected)</li><li>• Current &amp; Target Stock Levels</li><li>• Defect Rates</li><li>• Associated Keys: Supplier, Location, Logistics, Date</li></ul> | <ul style="list-style-type: none"><li>• <b>DimProduct:</b> SKU details, category, cost margins, style profile.</li><li>• <b>DimSupplier:</b> Supplier name, tier, performance tier, active contracts.</li><li>• <b>DimLocation:</b> Warehouse, manufacturing plant, regional retail hubs.</li><li>• <b>DimTransportation:</b> Modes (Air, Sea, Road, Rail), carriers, routes.</li><li>• <b>DimDate:</b> Dates, seasons, financial quarters, regional holidays.</li></ul> |

#### 3.2 Machine Learning Framework

To optimize resource constraints within a two-person development team, the engineering pipeline prioritizes a singular, high-impact predictive framework over multiple distributed lower-accuracy models.

- **Primary Target Framework** MVP Priority : **Stockout Risk Prediction Model**
  - **Objective:** Binary classification predicting whether a localized SKU will experience stock depletion within an operational lead-time window.
  - **Target Label:** Stockout Risk Status (High Risk / Low Risk).
  - **Key Features:** Stock levels, historical products sold, measured lead time, scheduled production volume, and calculated demand velocity.
- **Future Operational Pipeline Extensions:**
  - *Supplier Delay Framework:* Predicts logistics lag probabilities leveraging historical supplier performance, chosen route, and transportation mode.
  - *Defect Mitigation Model:* Pinpoints high-risk manufacturing anomalies using production volume, structural raw material inputs, and manufacturing lead times.

### 3.3 Business Intelligence & Analytics Dashboards

The analytical layer consolidates localized data schemas into four role-specific Power BI report interfaces:

- **Executive Performance Control:** Summary indicators charting overall corporate Revenue, Comprehensive Costs, Net Profit Margin, Systemic Average Lead Time, Defect Rates, and Baseline Stock Availability.
- **Inventory Management Intelligence:** Granular views highlighting high-risk SKUs, dynamic safety stock boundaries, active demand trends, and precise inventory turnover frequencies.
- **Supplier Lifecycle Analytics:** Aggregated data tracking individual vendor SLA compliance, variance in lead times, and specific defect rates.
- **Logistics & Transport Optimization:** Cost-benefit breakdowns of shipping costs, distribution of transportation modes, transit route efficiency, and individual carrier compliance metrics.

## 4. PROJECT SCHEDULE & MVP MILESTONES

The timeline is constrained to a rigorous 4-week sprints-based format executed by a localized 2-person delivery framework:

| PHASE                | TIMELINE    | CORE TECHNICAL ACTIVITIES                                                                                                                           | KEY MILESTONES                                  |
|----------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Phase 1: Foundations | Weeks 1 - 2 | Exploratory data profiling, ingestion orchestration, structuring standard Medallion architecture paths, formatting Gold star schema structures.     | Functional ETL and verified SQL data warehouse. |
| Phase 2: Modeling    | Week 3      | Feature engineering from Gold analytics tables, model building, cross-validation, hyperparameter tuning, risk classification scoring.               | Operational Stockout Prediction Model.          |
| Phase 3: Insights    | Week 4      | Establishing unified Power BI semantic connections, building and embedding visualizations, configuring risk thresholds, publishing report packages. | Live Enterprise Dashboard suite.                |